THE SYMBOL OF A ONE DIMENSIONAL MODULE IS THE PRODUCT OF THE SYMBOLS OF ITS MINIMAL PRIME QUOTIENTS RAISED TO THE LOCAL MULTIPLICITIES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 02Q5 (sources/stacks-chow/pinned-original-source.tex line 17577).

Allowed standard prerequisites (editorial).

- Existence of a filtration of a finite module over a Noetherian ring with prime quotients. Algebra, Lemma labelled algebra-lemma-filter-Noetherian-module (Stacks tag 00L0), invoked at proof lines 17598-17600. Supplies the filtration whose successive quotients are A/p_j, which is the backbone of the induction.
- The number of times a minimal prime of the support occurs in such a filtration equals the length of the localisation. Algebra, Lemma labelled algebra-lemma-filter-minimal-primes-in-support (Stacks tag 00L7), invoked at proof lines 17604-17606. Supplies the identification of the multiplicity with the count of occurrences, which is what makes the exponent in the statement appear.
- Multiplicativity of the symbol in a short exact sequence and comparison across a quotient. Same chapter, Lemmas labelled lemma-symbol-short-exact-sequence (source line 17503) and lemma-symbol-compare-modules (source line 17533), invoked at proof lines 17619-17620. Supply the two recurrence steps of the induction, for a minimal prime quotient and for the residue field quotient respectively.

Source reference. Original lemma 00L0: sources/stacks-chow/tags.txt, source lines 770–770. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 17600 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 00L7: sources/stacks-chow/tags.txt, source lines 777–777. Cross-chapter standard prerequisite located in the project tag list staged

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in the local snapshot; invoked at proof line 17606 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 02Q3: sources/stacks-chow/pinned-original-source.tex, source lines 17503–17503. Same-chapter pointer into the author environment lemma at source line 17503; invoked at proof line 17619 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 02Q4: sources/stacks-chow/pinned-original-source.tex, source lines 17533–17533. Same-chapter pointer into the author environment lemma at source line 17533; invoked at proof line 17620 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Let A be a Noetherian local ring. Let M be a finite A-module with $\dim(Supp(M)) = 1$. Let $a, b \in A$ nonzerodivisors on M. Let $\mathfrak{q}_1, \ldots, \mathfrak{q}_t$ be the minimal primes in the support of M. Then*

$$d_M(a,b) = \prod\nolimits_{i=1,\ldots,t} d_{A/\mathfrak{q}_i}(a,b)^{length_{A_{\mathfrak{q}_i}}(M_{\mathfrak{q}_i})}$$

as elements of κ^*.

Proof. Choose a filtration by A-submodules

$$0 = M_0 \subset M_1 \subset \ldots \subset M_n = M$$

such that each quotient M_j/M_{j-1} is isomorphic to $A/\mathfrak{p}_j$ for some prime ideal $\mathfrak{p}_j$ of A. See Algebra, Lemma 00L0. For each j we have either $\mathfrak{p}_j = \mathfrak{q}_i$ for some i, or $\mathfrak{p}_j = \mathfrak{m}_A$. Moreover, for a fixed i, the number of j such that $\mathfrak{p}_j = \mathfrak{q}_i$ is equal to $\text{length}_{A_{\mathfrak{q}_i}}(M_{\mathfrak{q}_i})$ by Algebra, Lemma 00L7. Hence $d_{M_j}(a,b)$ is defined for each j and

$$d_{M_j}(a,b) = \begin{cases} d_{M_{j-1}}(a,b) d_{A/\mathfrak{q}_i}(a,b) & \text{if} \quad \mathfrak{p}_j = \mathfrak{q}_i \\ d_{M_{j-1}}(a,b) & \text{if} \quad \mathfrak{p}_j = \mathfrak{m}_A \end{cases}$$

by Lemma 02Q3 in the first instance and Lemma 02Q4 in the second. Hence the lemma. □